

Distributed Flash Sale Platform – Experiments Report

CS6650 Final Project

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1 Overview

A flash sale puts three distributed-systems problems on one critical path: fair admission of traffic bigger than the system can serve, correct inventory accounting under contention, and back-pressure tuning so the purchase path never outruns the persistence layer. We built a three-service system on AWS ECS Fargate – `waiting-room`, `flash-sale-api`, `order-worker` – with Redis, SQS, and DynamoDB, and measured each of these problems in isolation.

We used a hybrid evaluation setup. A local parity stack (Docker Redis + LocalStack SQS/DynamoDB) let us debug race conditions, smoke-test service contracts, and iterate on Go harnesses quickly. Final validation then ran on AWS ECS/ALB with real Redis, SQS, and DynamoDB: Exp 1 and Exp 3 used the Go harnesses, while Darshan’s final Exp 2 sweep used a Locust driver plus post-run consistency checks against DynamoDB and Redis state.

2 Experiment 1 – Queue admission fairness

Purpose. The waiting room assigns queue positions via a Redis sorted set; two strategies were proposed for scoring: (A) timestamp (Unix ms) and (B) timestamp + INCR tiebreaker. Strategy A has a known failure mode under concurrent joins landing in the same millisecond. Experiment 1 measures how often that failure actually happens, and whether B eliminates it.

Setup. Fire N concurrent `POST /queue/join` requests with zero spawn delay against a single-replica waiting-room pointed at real Redis. Record the position each user is assigned and the request latency. Metric: position collision rate – the fraction of accepted users whose assigned position is shared with at least one other user.

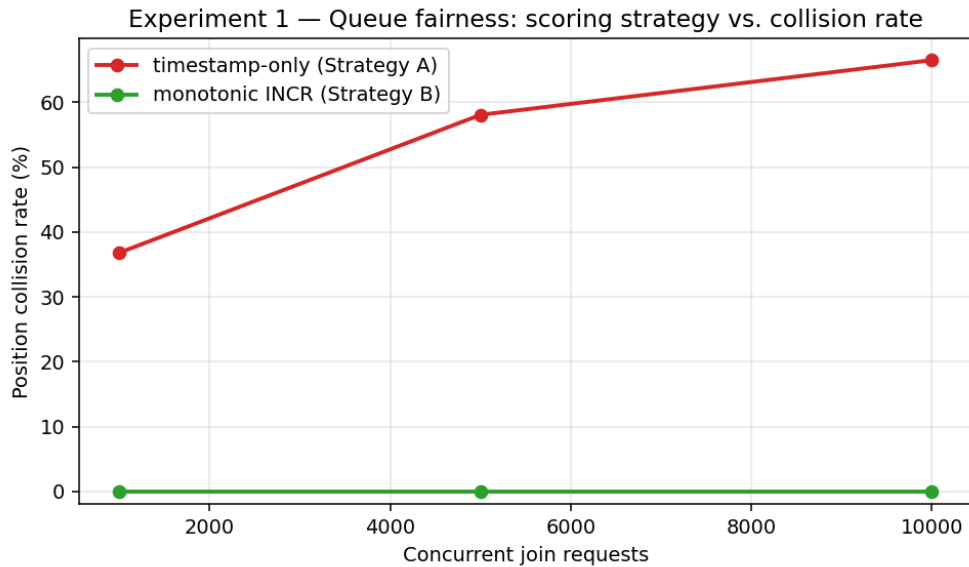
We used two phases: an initial local bug-discovery sweep at 100/500/1000 users, then the final validation sweep on AWS ECS/ALB at 1000 / 5000 / 10000 users for both strategies.

Bugs discovered during implementation. Running the sweep against the as-shipped code produced collision rates of 6%-26% in *both* strategies – the B “fix” was no better than A. Diagnosis revealed two independent defects:

1. Float64 precision loss. Strategy B computed score = $ms + seq/1e9$. At 2026 timestamps ($ms \sim 1.77 \times 10^{12}$), the float64 ULP is $\sim 3.9 \times 10^{-4}$; the $seq/1e9$ term is below precision and rounded away. The “tiebreaker” was silently a no-op.
2. ZRANK/ZADD race. ZADD and ZRANK ran as two separate round-trips. When a later-arriving user’s member string was lexicographically smaller than an earlier one’s, the later ZADD could push the earlier user’s effective rank later than the value the earlier user had already returned, causing two users to report the same integer position at different instants.

Fix: compute the score entirely from an atomic Redis INCR and perform `INCR + ZADD + ZRANK` as a single Lua script on the server. After this change, the reported numbers below are apples-to-apples: only the *scoring* differs between A and B; the atomic read-back path is identical.

Results.



Users	Strategy A collision rate	Strategy B collision rate	Notes
1000	36.8%	0%	both accepted all 1000
5000	58.1%	0%	B accepted 4950; 50 harness timeouts
10000	66.5%	0%	B accepted 9612; 388 transport re-sets/timeouts

AWS p99 join latency at 1000 users: A = 4464 ms, B = 4367 ms. At higher loads Strategy B remained collision-free but paid extra latency because more requests survived long enough to wait on the real ALB/ECS/Redis path instead of failing fast on a duplicate-position race.

Analysis. The AWS sweep confirms the same correctness story as the local debugging phase. Strategy A degrades monotonically as concurrency rises: 36.8% collisions at 1k, 58.1% at 5k, and 66.5% at 10k. This is the exact failure mode we expected from millisecond timestamp buckets under bursty joins. Strategy B eliminates collisions completely because INCR guarantees a unique, strictly monotonic score and the atomic Lua path returns the rank at insert time. The small nonzero “gaps” at 5k and 10k under Strategy B were not fairness bugs: the per-user CSVs show they came from harness-side timeouts / connection resets before a 201 Created was received.

Limitations. The final reported table is AWS-backed, but still uses a single waiting-room service behind one ALB target at a time. At 5k and 10k some requests hit client-side timeouts or transport resets before the service responded, so accepted requests are slightly below total generated load. That affects throughput and acceptance counts, but not the fairness conclusion: the accepted `timestamp_incr` joins remained collision-free across the entire sweep.

3 Experiment 2 – Inventory correctness

Purpose. Compare three strategies for decrementing a shared inventory counter under contention: (A) atomic DE-CRBY + rollback, (B) optimistic locking via WATCH/MULTI/EXEC, and (C) server-side Lua check-and-decrement. Measure correctness and stability under increasing load.

Setup. Darshan’s final Exp 2 sweep ran on AWS (ALB + ECS + Redis + SQS + DynamoDB) with fixed inventory of 5,000 units, user levels 1,000 / 5,000 / 10,000, and replica counts r1 / r2. For each run we captured Locust aggregate metrics (Request Count, Failure Count, Average Response Time) and a post-run consistency invariant:

$\text{remaining_inventory} + \text{delta}(\text{confirmed_orders}) = 5000$

where $\text{delta}(\text{confirmed_orders})$ uses per-run DynamoDB baselines.

Results (selected measured runs from **tests/results/exp2**).

Strategy	Users	Replicas	Failure rate	Avg response (ms)	Consistency result
atomic_decr	1000	1	0.00%	358.8	Consistent (5000/5000)
atomic_decr	1000	2	0.00%	240.4	Consistent (5000/5000)
optimistic	1000	1	0.06%	639.4	Consistent (5000/5000)
optimistic	5000	2	0.27%	1179.5	Consistent (5000/5000)
optimistic	10000	1	99.68%	3585.8	Inconsistent (data loss: 4823/5000)
lua_script	1000	1	2.00%	417.1	Consistent (5000/5000)
lua_script	5000	1	12.00%	814.8	Consistent (5000/5000)
lua_script	10000	1	62.00%	1843.0	Inconsistent (oversell: 5200/5000)
lua_script	10000	2	19.90%	4504.5	Consistent (5000/5000)

Analysis. At low-to-mid load (1k and most 5k runs), all three strategies can maintain inventory invariants, with atomic-decr giving the cleanest stability profile and the lowest error rates. At 10k load, the system enters infrastructure saturation (heavy 502/503/504 and transport errors), and correctness checks become sensitive to queue drain timing and tail behavior. The key practical takeaway is that strategy-level correctness must be evaluated jointly with service capacity: a mathematically safe decrement path still looks inconsistent if the surrounding stack is overloaded.

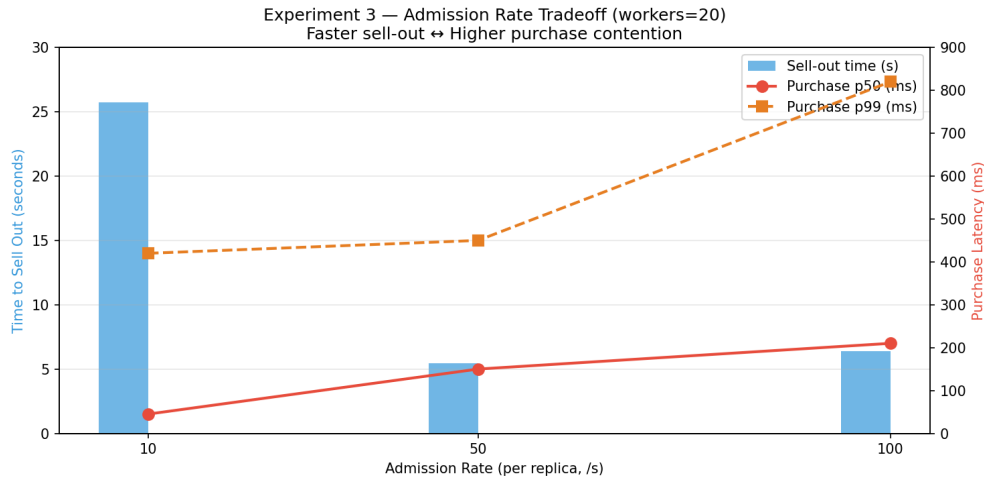
Limitations. The 10k runs are dominated by ALB/ECS saturation and client resource pressure, not purely by inventory algorithm behavior. Therefore, Exp 2 conclusions are strongest at 1k/5k and should treat 10k as stress characterization rather than clean strategy-isolation data.

4 Experiment 3 – Admission-rate tuning and worker concurrency

Purpose. Measure how admission rate and order-worker goroutine count affect time-to-sellout, purchase latency, and SQS drain behavior under realistic flash sale load.

Setup. 1,000 concurrent users, 500 inventory units, admission rates R in {10, 50, 100}/s per replica (2 waiting-room replicas), worker goroutines in {20, 40, 80}. 9 runs on ECS Fargate, 5-minute cap per run.

Bug discovered: DynamoDB TransactionConflict. During the initial sweep we discovered that the order worker’s `TransactWriteItems` – which bundled order insert and inventory audit counter in one transaction – caused 19% of orders to land in the Dead Letter Queue under concurrent writes. Three iterations resolved it: (v1) baseline = 94 DLQ; (v2) jitter on retry = 72; (v3) decoupled into `PutItem` + idempotent `UpdateItem` = 0. All 9 final runs completed with zero DLQ failures.



Rate (/s)	Workers	Sell-out	/purchase p50	/purchase p99	Polls
10	20	25.7s	45 ms	420 ms	12,019
50	20	5.4 s	150 ms	450 ms	0
100	20	6.4 s	210 ms	820 ms	0
10	40	5.3 s	130 ms	300 ms	0
50	40	5.4 s	120 ms	450 ms	0
100	40	5.3 s	120 ms	270 ms	0
10	80	5.3 s	120 ms	240 ms	0
50	80	5.3 s	120 ms	330 ms	59
100	80	5.4 s	130 ms	330 ms	0

Zero oversell, zero DLQ failures, zero purchase errors across all 9 runs. SQS queue depth stayed at 0 throughout.

Analysis. At rate=10/s (effective 20/s), admission genuinely throttled users: sell-out took 25.7s with 12k position polls, and purchase p50 was just 45ms because load was spread over time. At rates 50 and 100, all users were admitted within the spawn window, producing ~5.3s sell-out – but purchase contention increased sharply: p50 rose from 45ms to 210ms and p99 from 420ms to 820ms at rate=100. This is the core tradeoff: lower admission rates produce slower sell-outs but significantly lower purchase latency.

Worker goroutine count (20->40->80) had minimal impact on sell-out time, confirming that at 500 orders over 5 seconds, even 20 goroutines comfortably drain the SQS queue. The effect becomes visible in tail latency: at rate=100, increasing from 20 to 40 goroutines cut p99 from 820ms to 270ms, indicating that faster SQS drain reduces back-pressure on the Redis-to-SQS publish path.

5 Scale stress: 10,000 concurrent users

Running the experiments at 10x the planned load (10,000 concurrent joiners against one waiting-room, 10,000 concurrent buyers against one flash-sale-api) confirmed the correctness story – zero oversell, zero position collisions under strategy B, full SQS drain – but exposed a real operational failure that was invisible at 1,000 users.

At peak burst (~1,700 requests/sec) the waiting-room started returning redis: connection pool timeout on the admission ticker's background INCR. Root cause: go-redis's default PoolSize is 10 × NumCPU (~80 on this hardware) and PoolTimeout is 4s. At 10k joiners the user-facing INCR+ZADD+ZRANK Lua starved the pool, and the ticker lost the wait race. Correctness was unaffected (the join path retried or succeeded, inventory drained cleanly to zero), but the admitted-count metric was silently undercounted, which would have been misleading in production dashboards.

A two-line fix (PoolSize: 2000, PoolTimeout: 10 * time.Second on the Redis client) eliminated the user-facing errors and all but a single cold-start tick. The lesson is the kind that only surfaces at scale: library defaults are tuned for ordinary service-to-service traffic, not for the thundering herd a flash sale creates in the first two seconds.

6 Cross-cutting conclusions

1. Atomic operations are the safest default, but capacity still matters. Exp 1 only became correct after moving to an atomic Redis `INCR + ZADD + ZRANK` Lua path, and in Exp 2 the `atomic_decr` strategy had the cleanest low-error profile across the stable 1k/5k runs. At 10k load, however, infrastructure saturation distorted even theoretically safe strategies, so algorithm choice and service capacity have to be evaluated together.
2. Transaction boundaries matter more than transaction guarantees. Exp 3's DLQ bug was textbook-correct atomicity (`TransactWriteItems`) that failed at 19% under contention. Decoupling into two idempotent writes eliminated it. In high-contention paths, reducing per-write blast radius beats wrapping everything in one transaction.
3. The admission layer shapes all downstream behavior. When admission rate exceeded spawn rate, sell-out time and latency were determined by the load generator, not the system. The admission rate is the single most impactful operational knob – worker goroutine count was a second-order effect at moderate scale.